

Tutorial 2_summary

Tutorial 2 — Systematic Error, Error Bands, Feedback & Statistical Analysis

Source: Tutorial 2_Solutions.pdf

Instructor: Qilong Zhong, MSc, GTIIT

Corresponds to: Lecture 2 (Chapter 3 of Bentley)

1. Basic Systematic Error — Three Assumptions of Ideal Elements

1.1 Three Assumptions

Assumption	Description
Perfectly linear	Strictly proportional relationship between input and output
Zero intercept	When input = 0, output = 0 (i.e., $a = 0$)
No environmental inputs	Not affected by temperature, humidity, etc.

1.2 Series System Error

For ideal elements: $O_i = K_i I_i$

The output of the previous element is the input for the next:

- Level 1: $O_1 = K_1 I$
- Level 2: $O_2 = K_2 O_1 = K_2 K_1 I$
- Level 3: $O_3 = K_3 O_2 = K_3 K_2 K_1 I$

$$K_{sys} = K_1 \times K_2 \times \dots \times K_n$$

$$E = O - I = (K_1 K_2 \dots K_n - 1)I$$

✓ When $K_{sys} = 1$, error = 0 — perfectly accurate.

2. Error Band Modeling and Propagation

2.1 What is an Error Band ($\pm h$)?

Manufacturers typically provide an "accuracy range" (e.g., $\pm 1\%$). This means the actual output is restricted within a band of total width $2h$ around the ideal linear path.

2.2 From "Range" to "Variance" — Uniform Distribution

Statistical assumption: Error is equally likely at any point within $\pm h \rightarrow$ **Rectangular (uniform) distribution.**

$$\sigma^2 = \frac{h^2}{3}$$

This transforms a manufacturer's $\pm h$ spec into a **variance** usable in error propagation.

2.3 Variance Propagation in Series Systems

Errors accumulate level by level and are **amplified by the sensitivity** K of each subsequent element:

$$\sigma_{out,i}^2 = (K_i \cdot \sigma_{in,i})^2 + \frac{h_i^2}{3}$$

where $\sigma_{in,i} = \sigma_{out,i-1}$ (previous stage's variance becomes the next stage's input variance).

▲ **Key:** The initial input (true physical quantity) usually has **no random error**: $\sigma_{in,1}^2 = 0$.

Also: Sensitivity K_1 of the first element **never amplifies its own error** — the output variance of the first stage is simply $h_1^2/3$.

3. High-Gain Negative Feedback & Error Reduction

3.1 The Pain Point of Open-Loop Systems

In a series (open-loop) system, **any** parameter drift in the forward path is directly amplified proportionally $\rightarrow$ massive measurement errors. Searching for "perfect elements" is prohibitively expensive.

3.2 The Solution: Negative Feedback

Core idea: "Pull back" the actual output and compare it with the intended input through subtraction. Use the resulting "error signal" to drive the system to self-correct.

3.3 Closed-Loop Transfer Function

$$\frac{O}{I} = \frac{K_{forward}}{1 + K_{forward} \cdot K_{feedback}}$$

3.4 High-Gain Approximation (The "Magic")

When $K_{forward} \cdot K_{feedback} \gg 1$:

$$\frac{O}{I} \approx \frac{1}{K_{feedback}}$$

✅ **The "Leverage" Effect:** Overall accuracy becomes **decoupled** from the forward path (which can now use cheap, drift-prone components) and depends **solely** on the stability of the feedback element. This dramatically improves immunity to interference.

4. Comprehensive Statistical Error Analysis — The "Dual-Track" Algorithm

4.1 Real-World Complexity

Real measurement systems are **non-linear** (quadratic terms, exponential terms, etc.) and simultaneously affected by cross-interference from multiple random variables.

4.2 Track 1: Calculating Mean Error $\bar{E}$ (Systematic Bias)

Rule: Ignore fluctuations, plug in **mean values**.

$$\bar{E} = \bar{O} - \text{True input } I$$

Substitute all nominal mean values step-by-step into the non-linear equations to calculate the final measured mean $\bar{O}$.

4.3 Track 2: Calculating Error Standard Deviation σ_E (Random Fluctuation)

Rule: Use partial derivatives as **weights** for variance propagation.

For a multi-variable function $y = f(x_1, x_2, \dots, x_n)$:

$$\sigma_y^2 = \sum_{i=1}^n \left(\frac{\partial f}{\partial x_i} \right)^2 \sigma_{x_i}^2$$

⚠ **Critical pitfall:** Partial derivatives must be evaluated at the **mean values** from Track 1!

4.4 Engineering Diagnosis & Optimization

Symptom	Cause	Fix
$\bar{E}$ is large	Systematic bias / software algorithm error	Modify inverse calculation formula in microcontroller; upgrade to more precise polynomial or lookup table
σ_E is large	Random fluctuation from environment	Identify largest variance term; add auxiliary sensors (temperature, pressure) for real-time dynamic compensation

✅ **Example:** If fluid density ρ is hard-coded as constant 1.2 but actually varies with temperature and pressure, install temperature + pressure sensors and use the ideal gas law $\rho = p/(RT)$ for **real-time density compensation**.

5. Key Formulas Reference

Concept	Formula
Series system sensitivity	$K_{sys} = K_1 K_2 \dots K_n$
System error	$E = (K_{sys} - 1)I$
Error band → variance	$\sigma^2 = h^2/3$
Variance propagation (series)	$\sigma_{out,i}^2 = (K_i \sigma_{in,i})^2 + h_i^2/3$
Closed-loop gain	$\frac{O}{I} = \frac{K_F}{1 + K_F K_B}$
High-gain approx	$O/I \approx 1/K_B$
Multi-variable variance	$\sigma_y^2 = \sum (\partial f / \partial x_i)^2 \sigma_{x_i}^2$

✅ **Key Takeaway:** The "Dual-Track" algorithm separates **systematic error** (mean shift → fix software/calibration) from **random error** (variance → fix hardware/dynamic compensation). High-gain negative feedback is the most powerful technique for making system accuracy depend on the stable feedback element rather than the imperfect forward path.